

Unit One Learning outcomes

By the end of this unit, students:

- understand how the kinetic particle model and thermodynamics concepts describe and explain heating processes (1.1)
- understand how the nuclear model of the atom explains radioactivity, fission, fusion and the properties of radioactive nuclides (1.2)
- understand how charge is involved in the transfer and transformation of energy in electrical circuits (1.3)
- understand how scientific models and theories have developed and are applied to improve existing, and develop new, technologies (1.4)
- use science inquiry skills to design, conduct and analyse safe and effective investigations into heating processes, nuclear physics and electrical circuits, and to communicate methods and findings (1.5)
- use algebraic and graphical representations to calculate, analyse and predict measurable quantities associated with heating processes, nuclear reactions and electrical circuits (1.6)
- evaluate, with reference to empirical evidence, claims about heating processes, nuclear reactions and electrical technologies (1.7)
- communicate physics understanding using qualitative and quantitative representations in appropriate modes and genres. (1.8)

Unit Two Learning outcomes

By the end of this unit, students:

- understand that Newton's Laws of Motion describe the relationship between the forces acting on an object and its motion (2.1)
- understand that waves transfer energy and that a wave model can be used to explain the behaviour of sound (2.2)
- understand how scientific models and theories have developed and are applied to improve existing, and develop new, technologies (2.3)
- use science inquiry skills to design, conduct and analyse safe and effective investigations into linear motion and wave phenomena, and to communicate methods and findings (2.4)
- use algebraic and graphical representations to calculate, analyse and predict measurable quantities associated with linear and wave motion (2.5)
- evaluate, with reference to evidence, claims about motion and sound related phenomena and associated technologies (2.6)
- communicate physics understanding using qualitative and quantitative representations in appropriate modes and genres. (2.7)

Programme of content delivery

Wk	Topic	Learning outcomes/ Resources	Assessments
-----SPRING TERM-----			
2	Skill building/Working in Physics	1.8, 2.7	
3	- units and measurement - SI units - Unit prefixes, changing between quantities - Scientific notation - Significant figures - Scientific method/investigation process - Error and uncertainty - Graphing skills	- Intro to Physics iBook - Nelson Ch 12 & 13	
4	Vectors	2.4, 2.5, 2.7	
5	<i>add and subtract vectors in two dimensions</i> - vector/scalar quantities - vector diagrams and representation - resolving vectors/calculating the resultant - addition of vectors in one plane - addition of vectors in two planes - drawing scale diagrams - algebraic resolution of vectors - subtraction of vectors - addition of opposite - conditions for equilibrium/ calculation of the equilibrant - components of vectors, calculations using components - drawing free body diagrams, point mass and forces - solving ramp problems	- Vectors booklet - Nelson Ch 8	VECTORS AND WORKING IN PHYSICS TEST
6	Motion and Forces	2.3, 2.4, 2.5, 2.6, 2.7	
7	<i>equations of motion</i> - definitions of distance, disp, speed, v, and a graphs of motion - derive the equations of motion	- Motion booklet #1 - Nelson Ch 7	
8	- calculations using $s = vt$: distance, speed and time - concept of average or constant		MEASURING g BY FALLING OBJECT INVESTIGATION

Wk	Topic	Learning outcomes/ Resources	Assessments
9	speed/velocity when using $s = v t$ - calculations using $s = v t$: displacement, velocity and time - concepts of instantaneous speed and acceleration - calculations using $v = u + a t$ - changing between m/s and km/h - negative acceleration - calculations using $s = ut + \frac{1}{2} a t^2$ - calculations using $v^2 = u^2 + 2 a s$ - solving overtaking/interval problems - concept of acceleration due to gravity near the earth's surface - calculations involving vertical motion - graphical analysis of motion - working with displacement-time graphs - working with velocity-time graphs - working with acceleration-time graphs		
-----SUMMER TERM-----			
1	<u>Motion and Forces</u>	2.1, 2.3, 2.4, 2.5, 2.6, 2.7	
2	<i>newton's laws of motion; law of conservation of momentum; law of conservation of energy</i>	- Motion booklet #2 - Nelson Ch 8 & 9	
3	- Newton's 1st Law - define and calculations of momentum - law of conservation of momentum, with calculations - Newton's 2nd Law		AIRBAGS: MOMENTUM AND IMPULSE EVALUATION AND ANALYSIS
4	- N's 2nd L and momentum - elastic and inelastic collisions - concept and calculations of impulse - Newton's 3rd Law - definitions and calculations of mass and weight using $F = m a$ - solving 'lift' problems - concepts of work and energy - interchangeability - calculations of $W = F s$, including angle/ component of 'useful' work - law of conservation of energy - definition of kinetic energy - calculations using $KE = \frac{1}{2} m v^2$ - definition of potential energy - calculations using $PE = m g h$ - interchange of KE and PE - concept and calculations of power, using $P = E/t$ - efficiency of machines - calculations of efficiency - graphical analysis		MOTION TEST

Wk	Topic	Learning outcomes/ Resources	Assessments
5	Heating and Cooling	1.1, 1.4, 1.5, 1.6, 1.7, 1.8	
6	<i>Kinetic particle model, internal energy, temperature, specific heat capacity, change of state and latent heat, heat transfer</i>	- Nelson Ch 1 & 2	
7			SPECIFIC HEAT CAPACITY EXPERIMENT (PRAC TEST)
8	- kinetic particle model of matter and the states of matter, assumptions of		
9	- internal energy of substances, phase changes		
	- energy definitions, heat energy		HEATING AND COOLING TEST
	- difference between heat and temperature		
	- measuring temperature		
	- concept of thermal equilibrium		
	- calculations using $Q_{\text{loss}} = Q_{\text{gain}}$		
	- heating and cooling curves		
	- concept of specific heat		
	- calculations of specific heat capacity		
	- concept of latent heat		
	- specific latent heat of fusion, specific latent heat of vaporisation		
	- concept of internal energy		
	- work: calculations using change $Q = Q_{\text{gain}} + \text{work done}$		
	- energy transfer models		
	- conduction		
	- convection		
	- radiation		
	- applications of C, C and R, including energy-efficient houses and maintaining body temperature		
	- calculations of efficiency of heating systems		
	- conversion of heat energy into other energy forms		
	- concept of absolute zero and superconductivity/superfluidity at low temperatures		
-----AUTUMN TERM-----			
1	-----STUDY-----		
2	-----EXAMS-----		
3	-----EXAMS-----		

Wk	Topic	Learning outcomes/ Resources	Assessments
4	Electricity <i>static and dynamic electricity, work and energy in electric circuits, electrostatic forces and fields, current, resistance and potential difference,</i>	1.3, 1.4, 1.5, 1.6, 1.7, 1.8 - Electricity booklet - Nelson Ch 5 & 6	
5	<i>Ohm's Law, circuit design</i>		
6	- measurement of current - measurement of charge - calculations of current and charge		SERIES AND PARALLEL CIRCUITS INVESTIGATION
7	using $q = I t$ - concept of the elementary charge		
8	- concept of work and potential energy in an electric field, definition of potential difference aka voltage - the electron volt, relationship with measurements in joules - concept and definition of electric power - calculations using $P = V I$ - concept of electrical energy, using relationship $E = P t$ - concept of joule heat - calculations using $H E = I^2 R t$ - domestic power, kilowatt hours - Ohm's law, definition and former $V = I R$ - factors affecting resistance of conductors - concept and calculations of resistivity - ohmic and non-ohmic conductors - series and parallel circuits - circuit diagrams - calculations in series and parallel circuits - combinations of series and parallel circuits, calculations - electrical safety devices		ELECTRICITY TEST

Wk	Topic	Learning outcomes/ Resources	Assessments
9	Waves	2.2, 2.3, 2.4, 2.5, 2.6, 2.7	
10	<i>waves transfer energy, mechanical waves, wave diagrams: period, amplitude, wavelength, frequency, velocity, reflection and refraction phenomena, standing waves and interference phenomena, resonance and natural frequency, intensity</i>	- Waves booklet - Nelson Ch 10 & 11	
11	<i>what is a wave? types of waves wave terminology basic wave equations mechanical waves wave fronts transverse and longitudinal waves electromagnetic waves sound waves displacement/distance and displacement/time wave graphs compressions and rarefactions things affecting the velocity of sound calculations of speed of sound sonic booms the Doppler effect</i>		LIGHT EXPERIMENT (PRAC TEST)
-----WINTER TERM-----			
1	- reflection of waves - laws of reflection, TIR - applications	2.2, 2.3, 2.4, 2.5, 2.6, 2.7	
2	- refraction of sound waves	- Waves booklet	
3	- refraction of light, Snell's law	- Nelson Ch 10 & 11	
4	- forced vibrations - concept of resonance and resonant frequency - interference of waves - look at interference patterns - nodes and antinodes (maxima and minima) - concept of beats and relationship with interference - standing waves in strings - standing waves in open and closed pipes - harmonics of standing waves - production of music, superposition of waves - concept of intensity and its inverse square relationship with distance		WAVES TEST

Wk	Topic	Learning outcomes/ Resources	Assessments
5	Atomic Physics	1.2, 1.4, 1.5, 1.6, 1.7, 1.8	
6	<i>nuclear model of the atom, nuclear strong force and stability, alpha, beta and gamma decay, half life, absorbed dose and dose equivalent, binding energy per nucleon, mass difference, artificial transmutation, fission and fusion</i>		FISSION AND FUSION EVALUATION AND ANALYSIS
7			
8			
9	- ionising and non-ionising radiation - background radiation - historical progression to the nuclear model of the atom - four fundamental forces, 2 nuclear forces - nuclides, atomic number and mass number - connections with the periodic table - concept of radioactive decay - types of decay: alpha, beta, gamma - properties of alpha, beta and gamma radiation: penetrative power, range, effect of electric field, effect of magnetic field - detection of radiation: cloud chambers, geiger-muller tube - concept and calculations of half life - production of new nuclides by artificial transmutation - applications e.g. nuclear medicine and imaging - concept of binding energy and mass defect - relationship between mass and energy, including $E = mc^2$ - calculations of nuclear binding energy in eV and J - concepts of fusion and fission - nuclear reactors, parts of, applications and dangers - The Manhattan Project - biological effects of radiation - calculation of absorbed dose and dose equivalent		NUCLEAR TEST
10	-----STUDY-----		
	-----EXAMS-----		
	-----EXAMS-----		

Assessment Structure

Assessment Type (from syllabus)	Assessment type weighting (from syllabus)	Assessment Task weighting	Assessment Task	Outcomes assessed	Date
Science Inquiry	30%	5%	Investigation: Measuring g by falling object	2.1, 2.4, 2.5, 2.6, 2.7	Spring Term, Week 9
		5%	Evaluation and Analysis: Airbags, Momentum and Impulse	2.1, 2.3, 2.6, 2.7	Summer Term, Week 3
		5%	Experiment: Specific Heat Capacity	1.1, 1.5, 1.6, 1.7, 1.8	Summer Term, Week 7
		5%	Investigation: Series and Parallel circuits	1.3, 1.5, 1.6, 1.7, 1.8	Autumn Term, Week 6
		5%	Experiment: Light	2.2, 2.4, 2.5, 2.6, 2.7	Autumn Term, Week 11
		5%	Evaluation and Analysis: Fission and Fusion	1.2, 1.4, 1.7, 1.8	Winter Term, Week 6
Test	30%	5%	Vectors and Working in Physics Test	2.5, 1.8, 2.7	Spring Term, Week 5
		5%	Motion Test	2.1, 2.5, 2.6, 2.7	Summer Term, Week 4
		5%	Heating and Cooling Test	1.1, 1.6, 1.7, 1.8	Summer Term, Week 9
		5%	Electricity Test	1.3, 1.6, 1.7, 1.8	Autumn Term, Week 8
		5%	Waves Test	2.2, 2.5, 2.6, 2.7	Winter Term, Week 4
		5%	Nuclear Test	1.2, 1.6, 1.7, 1.8	Winter Term, Week 9
Examination	40%	15%	Semester 1 Examination	1.1, 1.4, 1.5, 1.6, 1.7, 1.8, 2.1, 2.3, 2.4, 2.5, 2.6, 2.7	AUTUMN TERM EXAM PERIOD
		25%	Semester 2 Examination	ALL	WINTER/ SPRING TERM EXAM PERIOD